Algorithmic Game Theory

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Contents

1 Introduction and Examples **2**
1.1 Mechanism design and incentives 2
1.2 Selfish routing and Braess's paradox 2
1.3 Equilibrium and computation 3

Lecture 1: Introduction and Examples

1.1 Mechanism design and incentives

In a system with strategic participants, the rules influence behavior. In the 2012 Olympic badminton tournament, two teams had already qualified for the knockout stage, but winning their final group match would put them on a path to face a particularly strong opponent earlier. Both teams tried to lose and were disqualified. The example illustrates how a rule intended to organize competition can create incentives at odds with the desired behavior.

Mechanism design studies rules that produce desirable outcomes when participants pursue their own objectives. Applications include auctions, school assignments, and kidney exchange. Computational considerations add questions of efficiency, approximation, and guarantees that remain meaningful under varying assumptions about participants.

1.2 Selfish routing and Braess's paradox

Consider a unit mass of negligible drivers traveling from s to t. A road labeled x has travel time equal to the fraction of traffic using it; a road labeled 1 takes one hour regardless of traffic. Initially only the two outer routes in the diagram exist. If a fraction x uses the upper route, its time is $1 + x$ and the lower route's time is $1 + (1 - x)$. At equilibrium traffic splits evenly, so everyone takes $3/2$ hours.

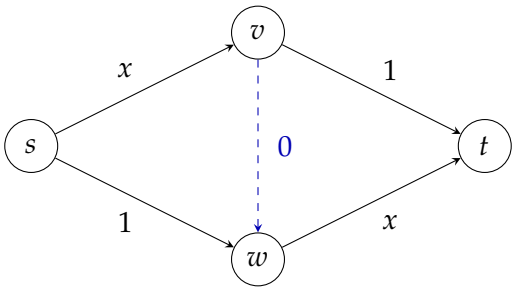


Figure 1: The dashed zero-time link is added to the original two-route network.

After adding a zero-time link $v \to w$, each driver can use the two congestible roads and avoid both constant-time roads. With all traffic on the new route, both congestible roads carry one unit, giving everyone a travel time of two hours. No individual driver can improve by switching to an outer route, which also takes two hours. Thus adding a road worsens the equilibrium for everyone: this is Braess's paradox.

A planner could instead split traffic over the outer routes, attaining average cost $3/2$ even with the new edge present. This is optimal: if a and b are the total flows on $s \to v$ and $w \to t$, respectively, then $a + b \geq 1$ and the total cost is $a^2 + b^2 + (1 - a) + (1 - b) \geq 3/2$. The ratio of equilibrium cost to optimal cost is therefore $2/(3/2) = 4/3$. The PRICE OF ANARCHY measures such inefficiency, usually using the worst equilibrium in a class.

A mechanical analogue replaces traffic by force and travel time by extension: cutting a taut connector can move two springs from bearing the full load in series to sharing it in parallel, raising the weight. The same network effect can arise without literal drivers.

1.3 Equilibrium and computation

A NASH EQUILIBRIUM is a strategy profile in which no player gains by changing strategy alone while the others remain fixed. Rock-paper-scissors has no pure equilibrium: some player can improve at every deterministic pair of choices. If both choose each move with probability $1/3$, every pure response has expected payoff zero, so neither can gain by deviating. This is a MIXED-STRATEGY equilibrium.

Every finite game has a mixed Nash equilibrium, but existence does not itself provide an efficient way to find one. Two-player zero-sum equilibria can be computed in polynomial time using linear programming; finding an equilibrium of a general two-player game is PPAD-complete. The computational difficulty raises a modeling question: when should an equilibrium be expected to arise from the limited computation or learning of the participants?

Source. Tim Roughgarden, Algorithmic Game Theory, Fall 2013, Lecture 1.